Model Comparison

Setup

```r
# install.packages(c("survey", "mgcv", "glmnet", "dplyr", "pROC", "ranger", "xgboost"))
library(survey)
```

```
Warning: package 'survey' was built under R version 4.5.2

Loading required package: grid

Loading required package: Matrix

Loading required package: survival


Attaching package: 'survey'

The following object is masked from 'package:graphics':

    dotchart
```

```r
library(mgcv)
```

```
Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.
```

```r
library(glmnet)
```

```
Loaded glmnet 4.1-10
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following object is masked from 'package:nlme':

    collapse

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(pROC)
```

```
Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

    cov, smooth, var
```

```r
library(ranger)
library(xgboost)
```

```
Warning: package 'xgboost' was built under R version 4.5.2
```

Data split and survey design

```
df <- readRDS("../data/brfss2024_clean.rds")

predictor_formula <- MEDCOST1 ~
  region +
  rurality +
  sex +
  agegroup +
  race +
  income +
  maritalstatus +
  education +
  employment +
  healthstatus +
  physhlth_days +
  menthlth_days +
  depression +
  chronic_count +
  any_disability +
  insurance +
  pers_doc +
  last_checkup

set.seed(263)
train_index <- unlist(lapply(
  split(seq_len(nrow(df)), df$MEDCOST1),
  function(idx) sample(idx, floor(0.8 * length(idx)))
))

train_df <- df[train_index, ]
test_df <- df[-train_index, ]

y_train <- train_df$MEDCOST1
y_test <- test_df$MEDCOST1

split_summary <- data.frame(
  Set = c("Training", "Testing"),
  N = c(nrow(train_df), nrow(test_df)),
  Event_Rate = round(c(mean(y_train), mean(y_test)), 3)
)
```

```r
split_summary
```

```
       Set      N Event_Rate
1 Training 302236       0.09
2  Testing  75559       0.09
```

```r
# create survey design object for weighted training models
options(survey.lonely.psu = "adjust")

brfss_train_design <- svydesign(
  id = ~PSU,
  strata = ~STSTR,
  weights = ~WEIGHT,
  data = train_df,
  nest = TRUE
)
```

Evaluation helpers

```r
safe_divide <- function(num, den) {
  ifelse(den == 0, NA_real_, num / den)
}

get_auc <- function(y, pred_prob) {
  roc_obj <- roc(y, as.numeric(pred_prob), quiet = TRUE)
  as.numeric(auc(roc_obj))
}

get_youden_threshold <- function(y, pred_prob) {
  roc_obj <- roc(y, as.numeric(pred_prob), quiet = TRUE)
  best <- coords(
    roc_obj,
    x = "best",
    best.method = "youden",
    ret = "threshold"
  )
  as.numeric(best[1])
}
```

```r
classification_metrics <- function(y, pred_prob, threshold) {
  pred_class <- ifelse(as.numeric(pred_prob) >= threshold, 1, 0)

  tp <- sum(pred_class == 1 & y == 1)
  tn <- sum(pred_class == 0 & y == 0)
  fp <- sum(pred_class == 1 & y == 0)
  fn <- sum(pred_class == 0 & y == 1)

  data.frame(
    Accuracy = mean(pred_class == y),
    Sensitivity = safe_divide(tp, tp + fn),
    Specificity = safe_divide(tn, tn + fp)
  )
}

evaluate_model <- function(model_name, train_y, train_prob, test_y, test_prob) {
  threshold <- get_youden_threshold(train_y, train_prob)
  metrics <- classification_metrics(test_y, test_prob, threshold)

  data.frame(
    Model = model_name,
    Test_AUC = get_auc(test_y, test_prob),
    Threshold = threshold,
    Accuracy = metrics$Accuracy,
    Sensitivity = metrics$Sensitivity,
    Specificity = metrics$Specificity
  )
}
```

Logistic Regression

Weighted

```r
weighted_glm <- svyglm(
  predictor_formula,
  design = brfss_train_design,
  family = quasibinomial()
)

summary(weighted_glm)
```

```
Call:
svyglm(formula = predictor_formula, design = brfss_train_design,
    family = quasibinomial())

Survey design:
svydesign(id = ~PSU, strata = ~STSTR, weights = ~WEIGHT, data = train_df,
    nest = TRUE)

Coefficients:
                                     Estimate Std. Error t value Pr(>|t|)
(Intercept)                        -4.6370113  0.0816589 -56.785  < 2e-16 ***
regionNortheast                    -0.1363063  0.0408467  -3.337 0.000847 ***
regionMidwest                      -0.2003320  0.0333426  -6.008 1.88e-09 ***
regionWest                         -0.1553013  0.0439711  -3.532 0.000413 ***
ruralityRural                      -0.0595700  0.0499737  -1.192 0.233252
sexFemale                           0.2336014  0.0316030   7.392 1.45e-13 ***
agegroup18-29                       0.4618200  0.0618224   7.470 8.03e-14 ***
agegroup30-39                       0.5575710  0.0593673   9.392  < 2e-16 ***
agegroup40-49                       0.5014442  0.0597400   8.394  < 2e-16 ***
agegroup50-59                       0.1885880  0.0566111   3.331 0.000865 ***
agegroup70+                        -0.7552759  0.0848425  -8.902  < 2e-16 ***
raceBlack_NH                        0.3214703  0.0470423   6.834 8.29e-12 ***
raceAsian_NH                        0.0606820  0.0938916   0.646 0.518087
raceAIAN_NH                         0.0072473  0.1308098   0.055 0.955817
raceHispanic                        0.3139616  0.0446818   7.027 2.12e-12 ***
raceOther_NH                        0.2847792  0.0662559   4.298 1.72e-05 ***
incomeUpper_mid                     0.7965609  0.0505694  15.752  < 2e-16 ***
incomeMiddle                        1.0750588  0.0538232  19.974  < 2e-16 ***
incomeLow_mid                       1.2320393  0.0709364  17.368  < 2e-16 ***
incomeLow                           1.0959523  0.0769951  14.234  < 2e-16 ***
incomeUnknown                       0.7282278  0.0598862  12.160  < 2e-16 ***
maritalstatusCurrently_not_married  0.0329601  0.0350691   0.940 0.347289
educationSome_college               0.1433122  0.0393034   3.646 0.000266 ***
educationHS_grad                    0.0001488  0.0423403   0.004 0.997196
educationLess_than_HS               0.0147860  0.0601614   0.246 0.805858
employmentUnemployed                0.1297953  0.0551859   2.352 0.018675 *
employmentNot_in_labor             -0.2380707  0.0580406  -4.102 4.10e-05 ***
employmentRetired                  -0.6834323  0.0811420  -8.423  < 2e-16 ***
employmentUnable_to_work           -0.4307711  0.0620440  -6.943 3.85e-12 ***
```

```
healthstatusVery_good              0.3131617  0.0578531   5.413 6.20e-08 ***
healthstatusGood                   0.5919055  0.0559314  10.583  < 2e-16 ***
healthstatusFair                   0.9707472  0.0633771  15.317  < 2e-16 ***
healthstatusPoor                   0.9876553  0.0898556  10.992  < 2e-16 ***
physhlth_days                      0.0152908  0.0018834   8.119 4.73e-16 ***
menthlth_days                      0.0273667  0.0016751  16.337  < 2e-16 ***
depression                         0.3010988  0.0364034   8.271  < 2e-16 ***
chronic_count                      0.0964308  0.0182342   5.288 1.23e-07 ***
any_disability                     0.5153845  0.0380575  13.542  < 2e-16 ***
insurancePrivate                   0.2468420  0.0564338   4.374 1.22e-05 ***
insuranceMedicare                 -0.2622653  0.0606624  -4.323 1.54e-05 ***
insuranceMedicaid_CHIP            -0.2213039  0.0562368  -3.935 8.31e-05 ***
insuranceMilitary_VA              -0.7928384  0.1307601  -6.063 1.34e-09 ***
insuranceOther_govt                0.2177498  0.0671705   3.242 0.001188 **
insuranceUninsured                 1.3722451  0.0522902  26.243  < 2e-16 ***
pers_docNo                         0.2380285  0.0449675   5.293 1.20e-07 ***
last_checkup1_2yrs                 0.4702457  0.0434283  10.828  < 2e-16 ***
last_checkup2_5yrs                 0.6747036  0.0565220  11.937  < 2e-16 ***
last_checkupOver_5yrs              0.5696645  0.0622011   9.158  < 2e-16 ***
last_checkupNever                  0.4700073  0.1773147   2.651 0.008033 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 0.9492774)

Number of Fisher Scoring iterations: 6
```

Unweighted

```r
unweighted_glm <- glm(
  predictor_formula,
  data = train_df,
  family = binomial()
)

summary(unweighted_glm)
```

```
Call:
glm(formula = predictor_formula, family = binomial(), data = train_df)
```

```
Coefficients:
                                    Estimate Std. Error  z value Pr(>|z|)    
(Intercept)                       -4.7264751  0.0415297 -113.809  < 2e-16 ***
regionNortheast                   -0.1499946  0.0213110   -7.038 1.95e-12 ***
regionMidwest                     -0.1811348  0.0194766   -9.300  < 2e-16 ***
regionWest                        -0.0947175  0.0203774   -4.648 3.35e-06 ***
ruralityRural                     -0.0577621  0.0220995   -2.614 0.008956 ** 
sexFemale                          0.2243000  0.0153449   14.617  < 2e-16 ***
agegroup18-29                      0.5897125  0.0292982   20.128  < 2e-16 ***
agegroup30-39                      0.6697934  0.0278624   24.039  < 2e-16 ***
agegroup40-49                      0.5468214  0.0271248   20.159  < 2e-16 ***
agegroup50-59                      0.2668021  0.0263404   10.129  < 2e-16 ***
agegroup70+                       -0.7425077  0.0340610  -21.799  < 2e-16 ***
raceBlack_NH                       0.3117026  0.0256580   12.148  < 2e-16 ***
raceAsian_NH                       0.0248702  0.0476954    0.521 0.602061    
raceAIAN_NH                        0.1712614  0.0502463    3.408 0.000653 ***
raceHispanic                       0.3383085  0.0217818   15.532  < 2e-16 ***
raceOther_NH                       0.2513350  0.0340068    7.391 1.46e-13 ***
incomeUpper_mid                    0.7538092  0.0251683   29.951  < 2e-16 ***
incomeMiddle                       1.1527285  0.0268630   42.911  < 2e-16 ***
incomeLow_mid                      1.2647825  0.0333850   37.885  < 2e-16 ***
incomeLow                          1.1278060  0.0377336   29.889  < 2e-16 ***
incomeUnknown                      0.7502955  0.0304723   24.622  < 2e-16 ***
maritalstatusCurrently_not_married 0.0355323  0.0163713    2.170 0.029977 *  
educationSome_college              0.0834646  0.0190874    4.373 1.23e-05 ***
educationHS_grad                  -0.0362193  0.0198820   -1.822 0.068498 .  
educationLess_than_HS              0.0338990  0.0294061    1.153 0.248998    
employmentUnemployed               0.1155862  0.0272385    4.243 2.20e-05 ***
employmentNot_in_labor            -0.2265222  0.0280272   -8.082 6.36e-16 ***
employmentRetired                 -0.7064575  0.0325262  -21.720  < 2e-16 ***
employmentUnable_to_work          -0.3983088  0.0297627  -13.383  < 2e-16 ***
healthstatusVery_good              0.2823574  0.0292240    9.662  < 2e-16 ***
healthstatusGood                   0.6272805  0.0279389   22.452  < 2e-16 ***
healthstatusFair                   0.9417902  0.0315466   29.854  < 2e-16 ***
healthstatusPoor                   1.0068199  0.0413771   24.333  < 2e-16 ***
physhlth_days                      0.0134359  0.0009015   14.904  < 2e-16 ***
menthlth_days                      0.0266038  0.0007926   33.566  < 2e-16 ***
depression                         0.2976400  0.0178513   16.673  < 2e-16 ***
chronic_count                      0.1000302  0.0071867   13.919  < 2e-16 ***
any_disability                     0.5243505  0.0177140   29.601  < 2e-16 ***
insurancePrivate                   0.2134449  0.0267084    7.992 1.33e-15 ***
insuranceMedicare                 -0.3197035  0.0281869  -11.342  < 2e-16 ***
```

```
insuranceMedicaid_CHIP             -0.3220330  0.0276946  -11.628  < 2e-16 ***
insuranceMilitary_VA               -0.9019072  0.0537913  -16.767  < 2e-16 ***
insuranceOther_govt                 0.1412721  0.0294061    4.804 1.55e-06 ***
insuranceUninsured                  1.3687649  0.0251772   54.365  < 2e-16 ***
pers_docNo                          0.2648656  0.0213840   12.386  < 2e-16 ***
last_checkup1_2yrs                  0.5667479  0.0221378   25.601  < 2e-16 ***
last_checkup2_5yrs                  0.6820424  0.0272971   24.986  < 2e-16 ***
last_checkupOver_5yrs               0.6064199  0.0303609   19.974  < 2e-16 ***
last_checkupNever                   0.3523421  0.0682454    5.163 2.43e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 183351  on 302235  degrees of freedom
Residual deviance: 137442  on 302187  degrees of freedom
AIC: 137540

Number of Fisher Scoring iterations: 6
```

Comparison

```r
# model summary comparison table
weighted_glm_summary <- summary(weighted_glm)$coefficients
unweighted_glm_summary <- summary(unweighted_glm)$coefficients

weighted_glm_df <- data.frame(
  Variable = rownames(weighted_glm_summary),
  Weighted_Estimate = weighted_glm_summary[, 1],
  Weighted_SE = weighted_glm_summary[, 2],
  Weighted_p = weighted_glm_summary[, 4]
)

unweighted_glm_df <- data.frame(
  Variable = rownames(unweighted_glm_summary),
  Unweighted_Estimate = unweighted_glm_summary[, 1],
  Unweighted_SE = unweighted_glm_summary[, 2],
  Unweighted_p = unweighted_glm_summary[, 4]
)

# OR and 95% CI
```

```r
weighted_glm_df$Weighted_OR <- exp(weighted_glm_df$Weighted_Estimate)
weighted_ci <- exp(confint(weighted_glm))
weighted_glm_df$Weighted_OR_lower <- weighted_ci[, 1]
weighted_glm_df$Weighted_OR_upper <- weighted_ci[, 2]

unweighted_glm_df$Unweighted_OR <- exp(unweighted_glm_df$Unweighted_Estimate)
unweighted_ci <- exp(confint.default(unweighted_glm))
unweighted_glm_df$Unweighted_OR_lower <- unweighted_ci[, 1]
unweighted_glm_df$Unweighted_OR_upper <- unweighted_ci[, 2]

comparison_glm_table <- merge(
  weighted_glm_df,
  unweighted_glm_df,
  by = "Variable"
)

comparison_glm_table <- comparison_glm_table %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

comparison_glm_table
```

```
                          Variable Weighted_Estimate Weighted_SE Weighted_p
1                      (Intercept)            -4.637       0.082      0.000
2                    agegroup18-29             0.462       0.062      0.000
3                    agegroup30-39             0.558       0.059      0.000
4                    agegroup40-49             0.501       0.060      0.000
5                    agegroup50-59             0.189       0.057      0.001
6                      agegroup70+            -0.755       0.085      0.000
7                   any_disability             0.515       0.038      0.000
8                    chronic_count             0.096       0.018      0.000
9                       depression             0.301       0.036      0.000
10                 educationHS_grad            0.000       0.042      0.997
11           educationLess_than_HS             0.015       0.060      0.806
12           educationSome_college             0.143       0.039      0.000
13           employmentNot_in_labor           -0.238       0.058      0.000
14                employmentRetired           -0.683       0.081      0.000
15          employmentUnable_to_work          -0.431       0.062      0.000
16             employmentUnemployed            0.130       0.055      0.019
17                healthstatusFair             0.971       0.063      0.000
18                healthstatusGood             0.592       0.056      0.000
19                healthstatusPoor             0.988       0.090      0.000
20           healthstatusVery_good             0.313       0.058      0.000
```

21	incomeLow	1.096	0.077	0.000
22	incomeLow_mid	1.232	0.071	0.000
23	incomeMiddle	1.075	0.054	0.000
24	incomeUnknown	0.728	0.060	0.000
25	incomeUpper_mid	0.797	0.051	0.000
26	insuranceMedicaid_CHIP	-0.221	0.056	0.000
27	insuranceMedicare	-0.262	0.061	0.000
28	insuranceMilitary_VA	-0.793	0.131	0.000
29	insuranceOther_govt	0.218	0.067	0.001
30	insurancePrivate	0.247	0.056	0.000
31	insuranceUninsured	1.372	0.052	0.000
32	last_checkup1_2yrs	0.470	0.043	0.000
33	last_checkup2_5yrs	0.675	0.057	0.000
34	last_checkupNever	0.470	0.177	0.008
35	last_checkupOver_5yrs	0.570	0.062	0.000
36	maritalstatusCurrently_not_married	0.033	0.035	0.347
37	menthlth_days	0.027	0.002	0.000
38	pers_docNo	0.238	0.045	0.000
39	physhlth_days	0.015	0.002	0.000
40	raceAIAN_NH	0.007	0.131	0.956
41	raceAsian_NH	0.061	0.094	0.518
42	raceBlack_NH	0.321	0.047	0.000
43	raceHispanic	0.314	0.045	0.000
44	raceOther_NH	0.285	0.066	0.000
45	regionMidwest	-0.200	0.033	0.000
46	regionNortheast	-0.136	0.041	0.001
47	regionWest	-0.155	0.044	0.000
48	ruralityRural	-0.060	0.050	0.233
49	sexFemale	0.234	0.032	0.000

	Weighted_OR	Weighted_OR_lower	Weighted_OR_upper	Unweighted_Estimate
1	0.010	0.008	0.011	-4.726
2	1.587	1.406	1.791	0.590
3	1.746	1.555	1.962	0.670
4	1.651	1.469	1.856	0.547
5	1.208	1.081	1.349	0.267
6	0.470	0.398	0.555	-0.743
7	1.674	1.554	1.804	0.524
8	1.101	1.063	1.141	0.100
9	1.351	1.258	1.451	0.298
10	1.000	0.921	1.087	-0.036
11	1.015	0.902	1.142	0.034
12	1.154	1.069	1.247	0.083
13	0.788	0.703	0.883	-0.227

14	0.505	0.431	0.592	-0.706
15	0.650	0.576	0.734	-0.398
16	1.139	1.022	1.269	0.116
17	2.640	2.332	2.989	0.942
18	1.807	1.620	2.017	0.627
19	2.685	2.251	3.202	1.007
20	1.368	1.221	1.532	0.282
21	2.992	2.573	3.479	1.128
22	3.428	2.983	3.940	1.265
23	2.930	2.637	3.256	1.153
24	2.071	1.842	2.329	0.750
25	2.218	2.009	2.449	0.754
26	0.801	0.718	0.895	-0.322
27	0.769	0.683	0.866	-0.320
28	0.453	0.350	0.585	-0.902
29	1.243	1.090	1.418	0.141
30	1.280	1.146	1.430	0.213
31	3.944	3.560	4.370	1.369
32	1.600	1.470	1.743	0.567
33	1.963	1.758	2.193	0.682
34	1.600	1.130	2.265	0.352
35	1.768	1.565	1.997	0.606
36	1.034	0.965	1.107	0.036
37	1.028	1.024	1.031	0.027
38	1.269	1.162	1.386	0.265
39	1.015	1.012	1.019	0.013
40	1.007	0.779	1.302	0.171
41	1.063	0.884	1.277	0.025
42	1.379	1.258	1.512	0.312
43	1.369	1.254	1.494	0.338
44	1.329	1.168	1.514	0.251
45	0.818	0.767	0.874	-0.181
46	0.873	0.805	0.945	-0.150
47	0.856	0.785	0.933	-0.095
48	0.942	0.854	1.039	-0.058
49	1.263	1.187	1.344	0.224

	Unweighted_SE	Unweighted_p	Unweighted_OR	Unweighted_OR_lower
1	0.042	0.000	0.009	0.008
2	0.029	0.000	1.803	1.703
3	0.028	0.000	1.954	1.850
4	0.027	0.000	1.728	1.638
5	0.026	0.000	1.306	1.240
6	0.034	0.000	0.476	0.445

7	0.018	0.000	1.689	1.632
8	0.007	0.000	1.105	1.090
9	0.018	0.000	1.347	1.300
10	0.020	0.068	0.964	0.928
11	0.029	0.249	1.034	0.977
12	0.019	0.000	1.087	1.047
13	0.028	0.000	0.797	0.755
14	0.033	0.000	0.493	0.463
15	0.030	0.000	0.671	0.633
16	0.027	0.000	1.123	1.064
17	0.032	0.000	2.565	2.411
18	0.028	0.000	1.873	1.773
19	0.041	0.000	2.737	2.524
20	0.029	0.000	1.326	1.252
21	0.038	0.000	3.089	2.869
22	0.033	0.000	3.542	3.318
23	0.027	0.000	3.167	3.004
24	0.030	0.000	2.118	1.995
25	0.025	0.000	2.125	2.023
26	0.028	0.000	0.725	0.686
27	0.028	0.000	0.726	0.687
28	0.054	0.000	0.406	0.365
29	0.029	0.000	1.152	1.087
30	0.027	0.000	1.238	1.175
31	0.025	0.000	3.930	3.741
32	0.022	0.000	1.763	1.688
33	0.027	0.000	1.978	1.875
34	0.068	0.000	1.422	1.244
35	0.030	0.000	1.834	1.728
36	0.016	0.030	1.036	1.003
37	0.001	0.000	1.027	1.025
38	0.021	0.000	1.303	1.250
39	0.001	0.000	1.014	1.012
40	0.050	0.001	1.187	1.075
41	0.048	0.602	1.025	0.934
42	0.026	0.000	1.366	1.299
43	0.022	0.000	1.403	1.344
44	0.034	0.000	1.286	1.203
45	0.019	0.000	0.834	0.803
46	0.021	0.000	0.861	0.826
47	0.020	0.000	0.910	0.874
48	0.022	0.009	0.944	0.904
49	0.015	0.000	1.251	1.214

	Unweighted_OR_upper
1	0.010
2	1.910
3	2.063
4	1.822
5	1.375
6	0.509
7	1.749
8	1.121
9	1.395
10	1.003
11	1.096
12	1.128
13	0.842
14	0.526
15	0.712
16	1.184
17	2.728
18	1.978
19	2.968
20	1.404
21	3.326
22	3.782
23	3.338
24	2.248
25	2.233
26	0.765
27	0.768
28	0.451
29	1.220
30	1.304
31	4.129
32	1.841
33	2.087
34	1.626
35	1.946
36	1.070
37	1.029
38	1.359
39	1.015
40	1.310
41	1.126
42	1.436

```
43                1.464
44                1.374
45                0.867
46                0.897
47                0.947
48                0.986
49                1.290
```

```
pred_train_w_glm <- as.numeric(predict(weighted_glm, type = "response"))
pred_test_w_glm <- as.numeric(predict(
  weighted_glm,
  newdata = test_df,
  type = "response"
))

pred_train_u_glm <- as.numeric(predict(unweighted_glm, type = "response"))
pred_test_u_glm <- as.numeric(predict(
  unweighted_glm,
  newdata = test_df,
  type = "response"
))
```

Logistic Regression with Lasso

```
x_train <- model.matrix(predictor_formula, data = train_df)[, -1]
x_test <- model.matrix(predictor_formula, data = test_df)[, -1]
```

Weighted

```
set.seed(263)
weighted_lasso <- cv.glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1,
  nfolds = 5,
  type.measure = "auc",
```

```
  weights = train_df$WEIGHT
)
```

Unweighted

```
set.seed(263)
unweighted_lasso <- cv.glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1,
  nfolds = 5,
  type.measure = "auc"
)
```

Comparison

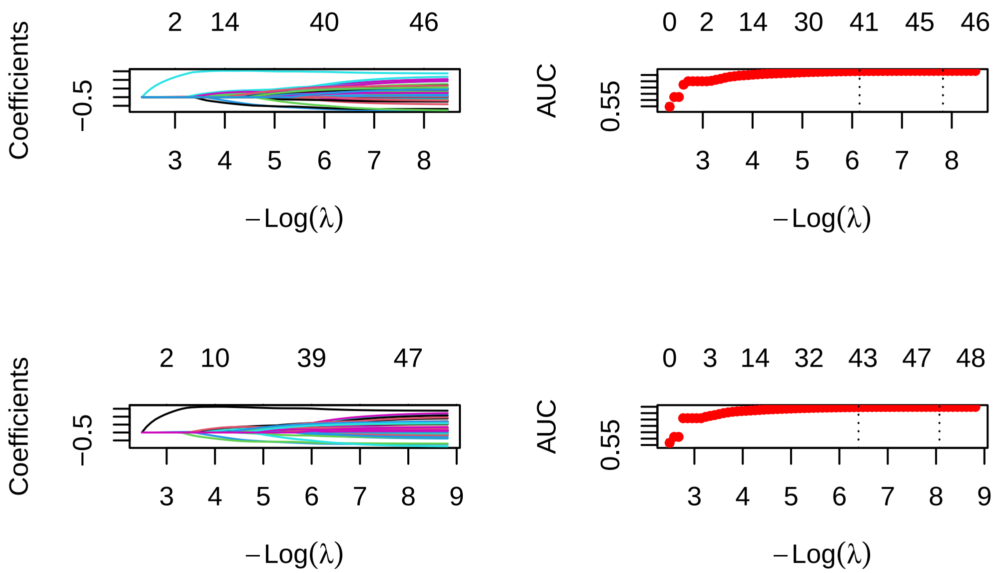
```
# Lasso selected variables
lasso_out <- cbind(
  coef(weighted_lasso, s = "lambda.1se"),
  coef(unweighted_lasso, s = "lambda.1se")
)
colnames(lasso_out) <- c("Weighted", "Unweighted")
lasso_out
```

```
49 x 2 sparse Matrix of class "dgCMatrix"
                                  Weighted  Unweighted
(Intercept)                    -3.84960928 -4.02266552
regionNortheast                -0.01361792 -0.02700524
regionMidwest                  -0.07221179 -0.04648782
regionWest                     -0.02822917  .
ruralityRural                   .           .
sexFemale                       0.17601392  0.17113155
agegroup18-29                   0.19092644  0.34854318
agegroup30-39                   0.27954769  0.41991106
agegroup40-49                   0.22557800  0.30240362
agegroup50-59                   .           0.04789947
agegroup70+                    -0.70181906 -0.71281730
```

```
raceBlack_NH                          0.24986261  0.26303677
raceAsian_NH                          .           .
raceAIAN_NH                           .           0.02699749
raceHispanic                          0.27975033  0.32761997
raceOther_NH                          0.16373612  0.15693613
incomeUpper_mid                       0.45134992  0.38931161
incomeMiddle                          0.70516453  0.74607267
incomeLow_mid                         0.79282341  0.78813728
incomeLow                             0.63401702  0.62264035
incomeUnknown                         0.30660651  0.29625374
maritalstatusCurrently_not_married    0.06773648  0.07215828
educationSome_college                 0.09074793  0.05945317
educationHS_grad                      .           .
educationLess_than_HS                 .           0.02297150
employmentUnemployed                  0.15564771  0.15564405
employmentNot_in_labor               -0.07653310 -0.05857963
employmentRetired                    -0.64363840 -0.66661076
employmentUnable_to_work             -0.23369290 -0.20193423
healthstatusVery_good                 .           .
healthstatusGood                      0.28456161  0.33125856
healthstatusFair                      0.65620981  0.63931822
healthstatusPoor                      0.61989177  0.65786712
physhlth_days                         0.01357281  0.01224454
menthlth_days                         0.02890296  0.02816446
depression                            0.28743583  0.28999731
chronic_count                         0.04697092  0.06122349
any_disability                        0.48809300  0.49333760
insurancePrivate                      0.19375060  0.16539052
insuranceMedicare                    -0.17649645 -0.24163014
insuranceMedicaid_CHIP               -0.01123660 -0.10490924
insuranceMilitary_VA                 -0.52751203 -0.62675740
insuranceOther_govt                   0.25200232  0.19334632
insuranceUninsured                    1.45892628  1.46012326
pers_docNo                            0.26273983  0.28898379
last_checkup1_2yrs                    0.38169559  0.47969162
last_checkup2_5yrs                    0.56958270  0.58453357
last_checkupOver_5yrs                 0.44492947  0.48686085
last_checkupNever                     0.25741087  0.17798610
```

```r
# Lasso plot
par(mfrow = c(2, 2))
plot(weighted_lasso$glmnet.fit, xvar = "lambda")
```

```r
plot(weighted_lasso)
plot(unweighted_lasso$glmnet.fit, xvar = "lambda")
plot(unweighted_lasso)
```




```r
pred_train_w_lasso <- as.numeric(predict(
  weighted_lasso,
  newx = x_train,
  s = "lambda.1se",
  type = "response"
))
pred_test_w_lasso <- as.numeric(predict(
  weighted_lasso,
  newx = x_test,
  s = "lambda.1se",
  type = "response"
))

pred_train_u_lasso <- as.numeric(predict(
  unweighted_lasso,
  newx = x_train,
  s = "lambda.1se",
  type = "response"
))
pred_test_u_lasso <- as.numeric(predict(
  unweighted_lasso,
```

```
  newx = x_test,
  s = "lambda.1se",
  type = "response"
))
```

GAM

Weighted

```
weighted_gam <- gam(
  MEDCOST1 ~
    s(physhlth_days) +
    s(menthlth_days) +
    region +
    rurality +
    sex +
    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  data = train_df,
  family = binomial(),
  weights = train_df$WEIGHT
)
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```r
summary(weighted_gam)
```

```
Family: binomial
Link function: logit

Formula:
MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +
    sex + agegroup + race + income + maritalstatus + education +
    employment + healthstatus + depression + chronic_count +
    any_disability + insurance + pers_doc + last_checkup

Parametric coefficients:
                                    Estimate Std. Error  z value Pr(>|z|)
(Intercept)                       -4.3708206  0.0015900 -2748.88   <2e-16 ***
regionNortheast                   -0.1454134  0.0008280  -175.61   <2e-16 ***
regionMidwest                     -0.2026572  0.0007429  -272.78   <2e-16 ***
regionWest                        -0.1646163  0.0007101  -231.81   <2e-16 ***
ruralityRural                     -0.0542352  0.0011434   -47.43   <2e-16 ***
sexFemale                          0.2098040  0.0005742   365.38   <2e-16 ***
agegroup18-29                      0.3703938  0.0011465   323.05   <2e-16 ***
agegroup30-39                      0.5056518  0.0011097   455.67   <2e-16 ***
agegroup40-49                      0.4744211  0.0010971   432.43   <2e-16 ***
agegroup50-59                      0.1844345  0.0010957   168.33   <2e-16 ***
agegroup70+                       -0.7401930  0.0015840  -467.29   <2e-16 ***
raceBlack_NH                       0.3257913  0.0008580   379.71   <2e-16 ***
raceAsian_NH                       0.0755378  0.0013285    56.86   <2e-16 ***
raceAIAN_NH                        0.0212827  0.0023158     9.19   <2e-16 ***
raceHispanic                       0.3263093  0.0007352   443.82   <2e-16 ***
raceOther_NH                       0.3006757  0.0012377   242.93   <2e-16 ***
incomeUpper_mid                    0.7971882  0.0009277   859.33   <2e-16 ***
incomeMiddle                       1.0858987  0.0009879  1099.22   <2e-16 ***
incomeLow_mid                      1.2583941  0.0012238  1028.29   <2e-16 ***
incomeLow                          1.1193561  0.0013781   812.23   <2e-16 ***
incomeUnknown                      0.7603928  0.0010949   694.50   <2e-16 ***
maritalstatusCurrently_not_married  0.0171194  0.0006161    27.79   <2e-16 ***
educationSome_college              0.1592516  0.0007687   207.17   <2e-16 ***
educationHS_grad                   0.0373577  0.0008110    46.06   <2e-16 ***
educationLess_than_HS              0.0586191  0.0010251    57.19   <2e-16 ***
employmentUnemployed               0.1243212  0.0009597   129.54   <2e-16 ***
employmentNot_in_labor            -0.2590539  0.0009265  -279.62   <2e-16 ***
employmentRetired                 -0.6612887  0.0014156  -467.13   <2e-16 ***
```

```
employmentUnable_to_work         -0.4204178  0.0011407  -368.57   <2e-16 ***
healthstatusVery_good             0.2487961  0.0010472   237.59   <2e-16 ***
healthstatusGood                  0.4954591  0.0009966   497.16   <2e-16 ***
healthstatusFair                  0.8241300  0.0011273   731.05   <2e-16 ***
healthstatusPoor                  0.9537749  0.0015182   628.22   <2e-16 ***
depression                        0.2275225  0.0006830   333.11   <2e-16 ***
chronic_count                     0.0862863  0.0002912   296.30   <2e-16 ***
any_disability                    0.4784376  0.0006542   731.36   <2e-16 ***
insurancePrivate                  0.2482076  0.0009650   257.20   <2e-16 ***
insuranceMedicare                -0.2476221  0.0011130  -222.48   <2e-16 ***
insuranceMedicaid_CHIP           -0.2004861  0.0010372  -193.30   <2e-16 ***
insuranceMilitary_VA             -0.7733336  0.0018528  -417.39   <2e-16 ***
insuranceOther_govt               0.2318134  0.0010486   221.07   <2e-16 ***
insuranceUninsured                1.3992627  0.0009040  1547.78   <2e-16 ***
pers_docNo                        0.2636427  0.0007517   350.72   <2e-16 ***
last_checkup1_2yrs                0.4706593  0.0008150   577.50   <2e-16 ***
last_checkup2_5yrs                0.6698035  0.0009670   692.68   <2e-16 ***
last_checkupOver_5yrs             0.5740601  0.0011048   519.59   <2e-16 ***
last_checkupNever                 0.5263804  0.0023726   221.86   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                   edf Ref.df  Chi.sq p-value
s(physhlth_days) 8.747  8.973  548819  <2e-16 ***
s(menthlth_days) 8.824  8.987 1040177  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.219   Deviance explained = 24.2%
UBRE = 313.85  Scale est. = 1         n = 302236
```

Unweighted

```r
unweighted_gam <- gam(
  MEDCOST1 ~
    s(physhlth_days) +
    s(menthlth_days) +
    region +
    rurality +
    sex +
```

```r
    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  data = train_df,
  family = binomial()
)

summary(unweighted_gam)
```

```
Family: binomial
Link function: logit

Formula:
MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +
    sex + agegroup + race + income + maritalstatus + education +
    employment + healthstatus + depression + chronic_count +
    any_disability + insurance + pers_doc + last_checkup

Parametric coefficients:
                                    Estimate Std. Error  z value Pr(>|z|)
(Intercept)                        -4.449756   0.041995 -105.958  < 2e-16 ***
regionNortheast                    -0.157795   0.021362   -7.387 1.51e-13 ***
regionMidwest                      -0.180624   0.019532   -9.247  < 2e-16 ***
regionWest                         -0.106331   0.020433   -5.204 1.95e-07 ***
ruralityRural                      -0.046363   0.022151   -2.093 0.036350 *
sexFemale                           0.193387   0.015413   12.547  < 2e-16 ***
agegroup18-29                       0.477097   0.029660   16.086  < 2e-16 ***
agegroup30-39                       0.595038   0.028071   21.198  < 2e-16 ***
agegroup40-49                       0.502310   0.027249   18.434  < 2e-16 ***
agegroup50-59                       0.245778   0.026419    9.303  < 2e-16 ***
agegroup70+                        -0.719457   0.034115  -21.089  < 2e-16 ***
```

```
raceBlack_NH                         0.320840   0.025723   12.473  < 2e-16 ***
raceAsian_NH                         0.052196   0.047885    1.090 0.275699
raceAIAN_NH                          0.180532   0.050321    3.588 0.000334 ***
raceHispanic                         0.366228   0.021852   16.759  < 2e-16 ***
raceOther_NH                         0.262439   0.034034    7.711 1.25e-14 ***
incomeUpper_mid                      0.751290   0.025221   29.789  < 2e-16 ***
incomeMiddle                         1.156102   0.026931   42.929  < 2e-16 ***
incomeLow_mid                        1.271483   0.033463   37.997  < 2e-16 ***
incomeLow                            1.143685   0.037785   30.268  < 2e-16 ***
incomeUnknown                        0.775775   0.030549   25.395  < 2e-16 ***
maritalstatusCurrently_not_married   0.021287   0.016436    1.295 0.195262
educationSome_college                0.103646   0.019152    5.412 6.24e-08 ***
educationHS_grad                     0.003871   0.019993    0.194 0.846481
educationLess_than_HS                0.084469   0.029582    2.855 0.004298 **
employmentUnemployed                 0.110712   0.027322    4.052 5.08e-05 ***
employmentNot_in_labor              -0.238979   0.028089   -8.508  < 2e-16 ***
employmentRetired                   -0.693713   0.032613  -21.271  < 2e-16 ***
employmentUnable_to_work            -0.380120   0.029717  -12.791  < 2e-16 ***
healthstatusVery_good                0.209140   0.029448    7.102 1.23e-12 ***
healthstatusGood                     0.514293   0.028309   18.167  < 2e-16 ***
healthstatusFair                     0.787034   0.032037   24.566  < 2e-16 ***
healthstatusPoor                     0.953753   0.041396   23.040  < 2e-16 ***
depression                           0.220052   0.017947   12.261  < 2e-16 ***
chronic_count                        0.088391   0.007203   12.271  < 2e-16 ***
any_disability                       0.490861   0.017743   27.665  < 2e-16 ***
insurancePrivate                     0.219553   0.026794    8.194 2.52e-16 ***
insuranceMedicare                   -0.305154   0.028250  -10.802  < 2e-16 ***
insuranceMedicaid_CHIP              -0.305355   0.027707  -11.021  < 2e-16 ***
insuranceMilitary_VA                -0.875437   0.053694  -16.304  < 2e-16 ***
insuranceOther_govt                  0.152086   0.029465    5.162 2.45e-07 ***
insuranceUninsured                   1.404363   0.025370   55.355  < 2e-16 ***
pers_docNo                           0.287362   0.021479   13.379  < 2e-16 ***
last_checkup1_2yrs                   0.563799   0.022203   25.393  < 2e-16 ***
last_checkup2_5yrs                   0.676201   0.027408   24.672  < 2e-16 ***
last_checkupOver_5yrs                0.612276   0.030514   20.066  < 2e-16 ***
last_checkupNever                    0.396017   0.068746    5.761 8.38e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                   edf Ref.df Chi.sq p-value
s(physhlth_days) 6.641  7.506  635.2  <2e-16 ***
s(menthlth_days) 8.285  8.801 1453.8  <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =   0.21   Deviance explained = 25.6%
UBRE = -0.54815  Scale est. = 1         n = 302236
```

Comparison

```r
# coefficients comparison table
weighted_gam_coef <- summary(weighted_gam)$p.coeff
unweighted_gam_coef <- summary(unweighted_gam)$p.coeff

weighted_gam_df <- data.frame(
  Variable = names(weighted_gam_coef),
  Weighted_Est = weighted_gam_coef,
  Weighted_OR = exp(weighted_gam_coef)
)

unweighted_gam_df <- data.frame(
  Variable = names(unweighted_gam_coef),
  Unweighted_Est = unweighted_gam_coef,
  Unweighted_OR = exp(unweighted_gam_coef)
)

gam_comparison <- merge(weighted_gam_df, unweighted_gam_df, by = "Variable")
gam_comparison <- gam_comparison %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))
gam_comparison
```

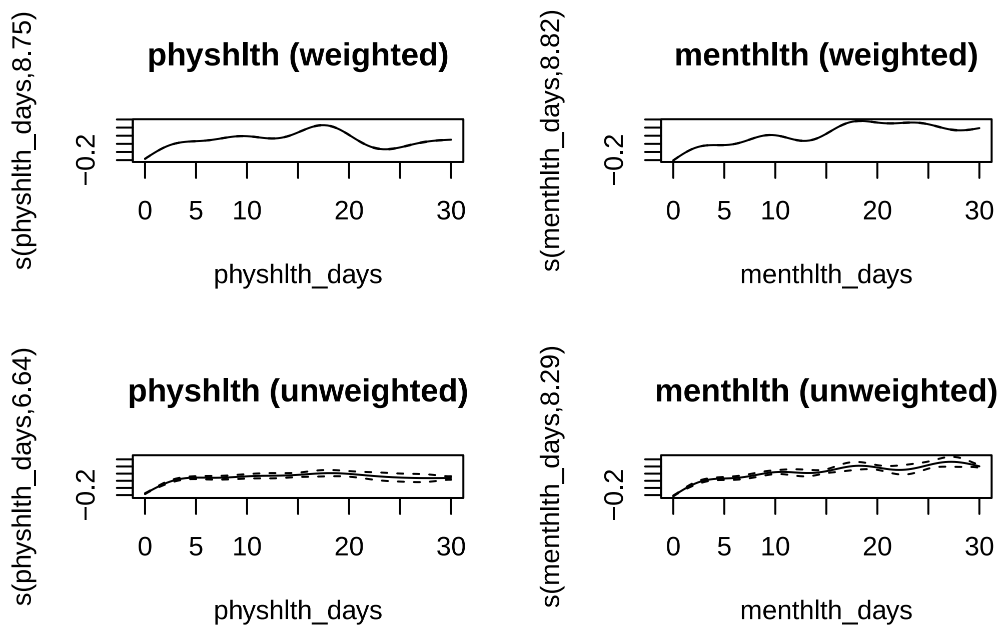
```
                          Variable Weighted_Est Weighted_OR Unweighted_Est
1                      (Intercept)       -4.371       0.013         -4.450
2                    agegroup18-29        0.370       1.448          0.477
3                    agegroup30-39        0.506       1.658          0.595
4                    agegroup40-49        0.474       1.607          0.502
5                    agegroup50-59        0.184       1.203          0.246
6                      agegroup70+       -0.740       0.477         -0.719
7                   any_disability        0.478       1.614          0.491
8                    chronic_count        0.086       1.090          0.088
9                       depression        0.228       1.255          0.220
10                 educationHS_grad        0.037       1.038          0.004
```

11	educationLess_than_HS	0.059	1.060	0.084
12	educationSome_college	0.159	1.173	0.104
13	employmentNot_in_labor	-0.259	0.772	-0.239
14	employmentRetired	-0.661	0.516	-0.694
15	employmentUnable_to_work	-0.420	0.657	-0.380
16	employmentUnemployed	0.124	1.132	0.111
17	healthstatusFair	0.824	2.280	0.787
18	healthstatusGood	0.495	1.641	0.514
19	healthstatusPoor	0.954	2.595	0.954
20	healthstatusVery_good	0.249	1.282	0.209
21	incomeLow	1.119	3.063	1.144
22	incomeLow_mid	1.258	3.520	1.271
23	incomeMiddle	1.086	2.962	1.156
24	incomeUnknown	0.760	2.139	0.776
25	incomeUpper_mid	0.797	2.219	0.751
26	insuranceMedicaid_CHIP	-0.200	0.818	-0.305
27	insuranceMedicare	-0.248	0.781	-0.305
28	insuranceMilitary_VA	-0.773	0.461	-0.875
29	insuranceOther_govt	0.232	1.261	0.152
30	insurancePrivate	0.248	1.282	0.220
31	insuranceUninsured	1.399	4.052	1.404
32	last_checkup1_2yrs	0.471	1.601	0.564
33	last_checkup2_5yrs	0.670	1.954	0.676
34	last_checkupNever	0.526	1.693	0.396
35	last_checkupOver_5yrs	0.574	1.775	0.612
36	maritalstatusCurrently_not_married	0.017	1.017	0.021
37	pers_docNo	0.264	1.302	0.287
38	raceAIAN_NH	0.021	1.022	0.181
39	raceAsian_NH	0.076	1.078	0.052
40	raceBlack_NH	0.326	1.385	0.321
41	raceHispanic	0.326	1.386	0.366
42	raceOther_NH	0.301	1.351	0.262
43	regionMidwest	-0.203	0.817	-0.181
44	regionNortheast	-0.145	0.865	-0.158
45	regionWest	-0.165	0.848	-0.106
46	ruralityRural	-0.054	0.947	-0.046
47	sexFemale	0.210	1.233	0.193

	Unweighted_OR
1	0.012
2	1.611
3	1.813
4	1.653
5	1.279

6	0.487
7	1.634
8	1.092
9	1.246
10	1.004
11	1.088
12	1.109
13	0.787
14	0.500
15	0.684
16	1.117
17	2.197
18	1.672
19	2.595
20	1.233
21	3.138
22	3.566
23	3.178
24	2.172
25	2.120
26	0.737
27	0.737
28	0.417
29	1.164
30	1.246
31	4.073
32	1.757
33	1.966
34	1.486
35	1.845
36	1.022
37	1.333
38	1.198
39	1.054
40	1.378
41	1.442
42	1.300
43	0.835
44	0.854
45	0.899
46	0.955
47	1.213

```
# smooth term plot
par(mfrow = c(2, 2))
plot(weighted_gam, select = 1, main = "physhlth (weighted)")
plot(weighted_gam, select = 2, main = "menthlth (weighted)")
plot(unweighted_gam, select = 1, main = "physhlth (unweighted)")
plot(unweighted_gam, select = 2, main = "menthlth (unweighted)")
```




```
# p-values for smooth terms
summary(weighted_gam)$s.pv
```

```
[1] 0 0
```

```
summary(unweighted_gam)$s.pv
```

```
[1] 0 0
```

```
pred_train_w_gam <- as.numeric(predict(weighted_gam, type = "response"))
pred_test_w_gam <- as.numeric(predict(
  weighted_gam,
  newdata = test_df,
  type = "response"
))
```

```r
pred_train_u_gam <- as.numeric(predict(unweighted_gam, type = "response"))
pred_test_u_gam <- as.numeric(predict(
  unweighted_gam,
  newdata = test_df,
  type = "response"
))
```

Random Forest

```r
rf_train_df <- train_df %>%
  mutate(MEDCOST1 = factor(MEDCOST1, levels = c(0, 1)))

rf_test_df <- test_df %>%
  mutate(MEDCOST1 = factor(MEDCOST1, levels = c(0, 1)))

rf_grid <- expand.grid(
  mtry = c(4, 6, 9),
  min.node.size = c(50, 100),
  KEEP.OUT.ATTRS = FALSE
)

set.seed(263)
rf_tuning <- lapply(seq_len(nrow(rf_grid)), function(i) {
  fit <- ranger(
    predictor_formula,
    data = rf_train_df,
    probability = TRUE,
    num.trees = 300,
    mtry = rf_grid$mtry[i],
    min.node.size = rf_grid$min.node.size[i],
    respect.unordered.factors = "order",
    seed = 263
  )

  data.frame(
    mtry = rf_grid$mtry[i],
    min.node.size = rf_grid$min.node.size[i],
    OOB_Error = fit$prediction.error
  )
```

```
}) %>%
  bind_rows() %>%
  arrange(OOB_Error)
```

```
Growing trees.. Progress: 64%. Estimated remaining time: 17 seconds.
Growing trees.. Progress: 53%. Estimated remaining time: 27 seconds.
Growing trees.. Progress: 40%. Estimated remaining time: 47 seconds.
Growing trees.. Progress: 79%. Estimated remaining time: 16 seconds.
Growing trees.. Progress: 72%. Estimated remaining time: 12 seconds.
Growing trees.. Progress: 54%. Estimated remaining time: 26 seconds.
Growing trees.. Progress: 41%. Estimated remaining time: 44 seconds.
Growing trees.. Progress: 83%. Estimated remaining time: 12 seconds.
```

```
rf_tuning
```

```
  mtry min.node.size  OOB_Error
1    4           100 0.06440932
2    6           100 0.06441033
3    4            50 0.06447070
4    6            50 0.06454268
5    9           100 0.06454389
6    9            50 0.06474709
```

```
best_rf <- rf_tuning[1, ]

set.seed(263)
random_forest <- ranger(
  predictor_formula,
  data = rf_train_df,
  probability = TRUE,
  num.trees = 500,
  mtry = best_rf$mtry,
  min.node.size = best_rf$min.node.size,
  importance = "impurity",
  respect.unordered.factors = "order",
  seed = 263
)
```

```
Growing trees.. Progress: 41%. Estimated remaining time: 44 seconds.
Growing trees.. Progress: 83%. Estimated remaining time: 13 seconds.
```

```r
pred_train_rf <- predict(random_forest, data = rf_train_df)$predictions[, "1"]
pred_test_rf <- predict(random_forest, data = rf_test_df)$predictions[, "1"]

rf_importance <- data.frame(
  Variable = names(random_forest$variable.importance),
  Importance = as.numeric(random_forest$variable.importance)
) %>%
  arrange(desc(Importance)) %>%
  mutate(Importance = round(Importance, 3))

head(rf_importance, 15)
```

```
         Variable Importance
1       insurance   3451.676
2   menthlth_days   1785.735
3        agegroup   1030.043
4   physhlth_days    976.965
5          income    968.388
6    healthstatus    957.686
7    last_checkup    807.454
8      employment    732.712
9        pers_doc    614.652
10 any_disability    534.360
11           race    457.123
12  chronic_count    433.469
13     depression    364.587
14      education    325.380
15         region    292.720
```

Boosting

```r
dtrain <- xgb.DMatrix(data = x_train, label = y_train)
dtest <- xgb.DMatrix(data = x_test, label = y_test)

xgb_grid <- expand.grid(
  eta = c(0.05, 0.10),
  max_depth = c(2, 3),
  KEEP.OUT.ATTRS = FALSE
)
```

```r
scale_pos_weight <- sum(y_train == 0) / sum(y_train == 1)

set.seed(263)
xgb_tuning <- lapply(seq_len(nrow(xgb_grid)), function(i) {
  params <- list(
    objective = "binary:logistic",
    eval_metric = "auc",
    eta = xgb_grid$eta[i],
    max_depth = xgb_grid$max_depth[i],
    min_child_weight = 5,
    subsample = 0.8,
    colsample_bytree = 0.8,
    scale_pos_weight = scale_pos_weight
  )

  cv_fit <- xgb.cv(
    params = params,
    data = dtrain,
    nrounds = 200,
    nfold = 3,
    early_stopping_rounds = 10,
    verbose = 0
  )

  cv_results <- cv_fit$evaluation_log
  best_row <- which.max(cv_results$test_auc_mean)

  data.frame(
    eta = xgb_grid$eta[i],
    max_depth = xgb_grid$max_depth[i],
    best_iteration = cv_results$iter[best_row],
    CV_AUC = cv_results$test_auc_mean[best_row]
  )
}) %>%
  bind_rows() %>%
  arrange(desc(CV_AUC))

xgb_tuning
```

```
   eta max_depth best_iteration    CV_AUC
1 0.10         3            200 0.8539367
```

```
2 0.10         2            200 0.8497087
3 0.05         3            200 0.8495945
4 0.05         2            200 0.8433611
```

```
best_xgb <- xgb_tuning[1, ]

xgb_params <- list(
  objective = "binary:logistic",
  eval_metric = "auc",
  eta = best_xgb$eta,
  max_depth = as.integer(best_xgb$max_depth),
  min_child_weight = 5,
  subsample = 0.8,
  colsample_bytree = 0.8,
  scale_pos_weight = scale_pos_weight
)

set.seed(263)
xgb_model <- xgb.train(
  params = xgb_params,
  data = dtrain,
  nrounds = as.integer(best_xgb$best_iteration),
  verbose = 0
)

pred_train_xgb <- predict(xgb_model, dtrain)
pred_test_xgb <- predict(xgb_model, dtest)

xgb_importance <- xgb.importance(
  feature_names = colnames(x_train),
  model = xgb_model
) %>%
  as.data.frame()

head(xgb_importance, 15)
```

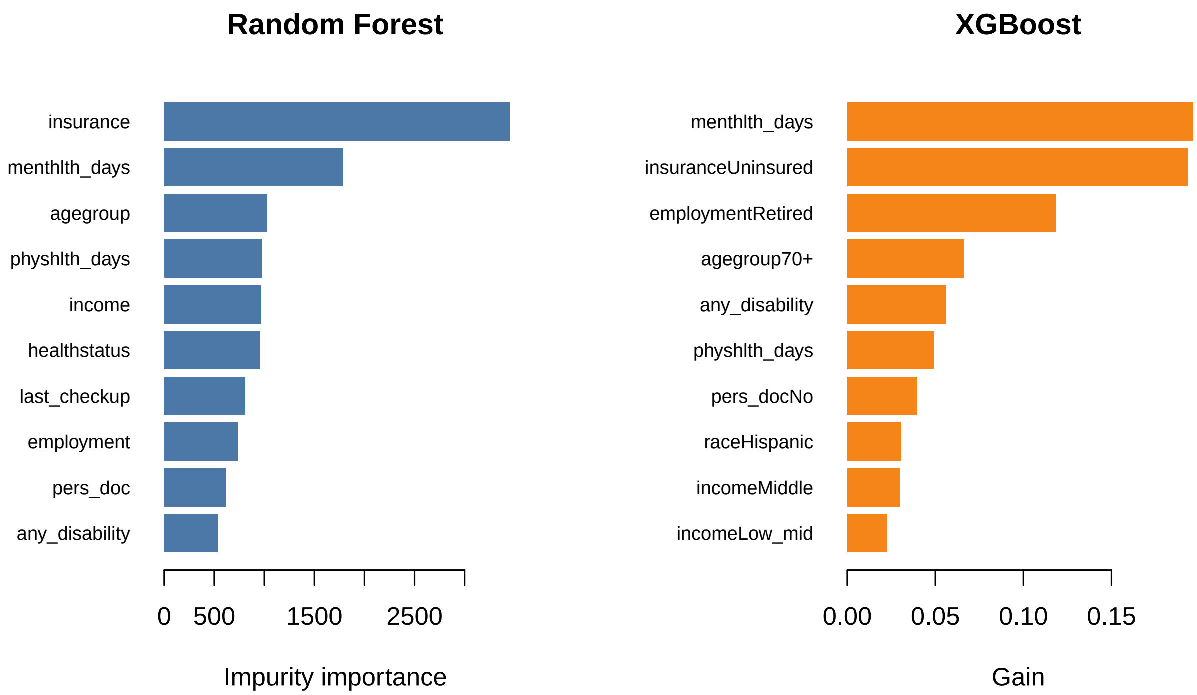
```
                          Feature       Gain      Cover  Frequency
1                    menthlth_days 0.19621177 0.04401531 0.05785714
2             insuranceUninsured 0.19304188 0.04073688 0.02500000
3              employmentRetired 0.11836181 0.02240954 0.02428571
4                     agegroup70+ 0.06628992 0.02898012 0.03714286
5                  any_disability 0.05620225 0.02577227 0.02428571
```

```
6                         physhlth_days 0.04923127 0.02947217 0.05142857
7                            pers_docNo 0.03944139 0.01663315 0.02714286
8                          raceHispanic 0.03051371 0.01811074 0.01357143
9                          incomeMiddle 0.03004036 0.06244526 0.04000000
10                        incomeLow_mid 0.02247781 0.04876644 0.03357143
11 maritalstatusCurrently_not_married 0.01733071 0.01613894 0.03214286
12                    healthstatusFair 0.01562171 0.02198432 0.01714286
13                           depression 0.01290773 0.01657696 0.02142857
14                      incomeUpper_mid 0.01174884 0.04549752 0.04071429
15                            incomeLow 0.01096394 0.03491154 0.02857143
```

Tree-based model figures

```r
#png(filename = "../figures/tree_model_variable_importance.png", width = 1200, height = 650,

top_rf <- head(rf_importance, 10)
top_xgb <- head(xgb_importance, 10)

old_par <- par(no.readonly = TRUE)
par(mfrow = c(1, 2), mar = c(5, 10, 4, 1))

barplot(
  rev(top_rf$Importance),
  names.arg = rev(top_rf$Variable),
  horiz = TRUE,
  las = 1,
  cex.names = 0.75,
  main = "Random Forest",
  xlab = "Impurity importance",
  col = "#4C78A8",
  border = NA
)

barplot(
  rev(top_xgb$Gain),
  names.arg = rev(top_xgb$Feature),
  horiz = TRUE,
  las = 1,
  cex.names = 0.75,
  main = "XGBoost",
```

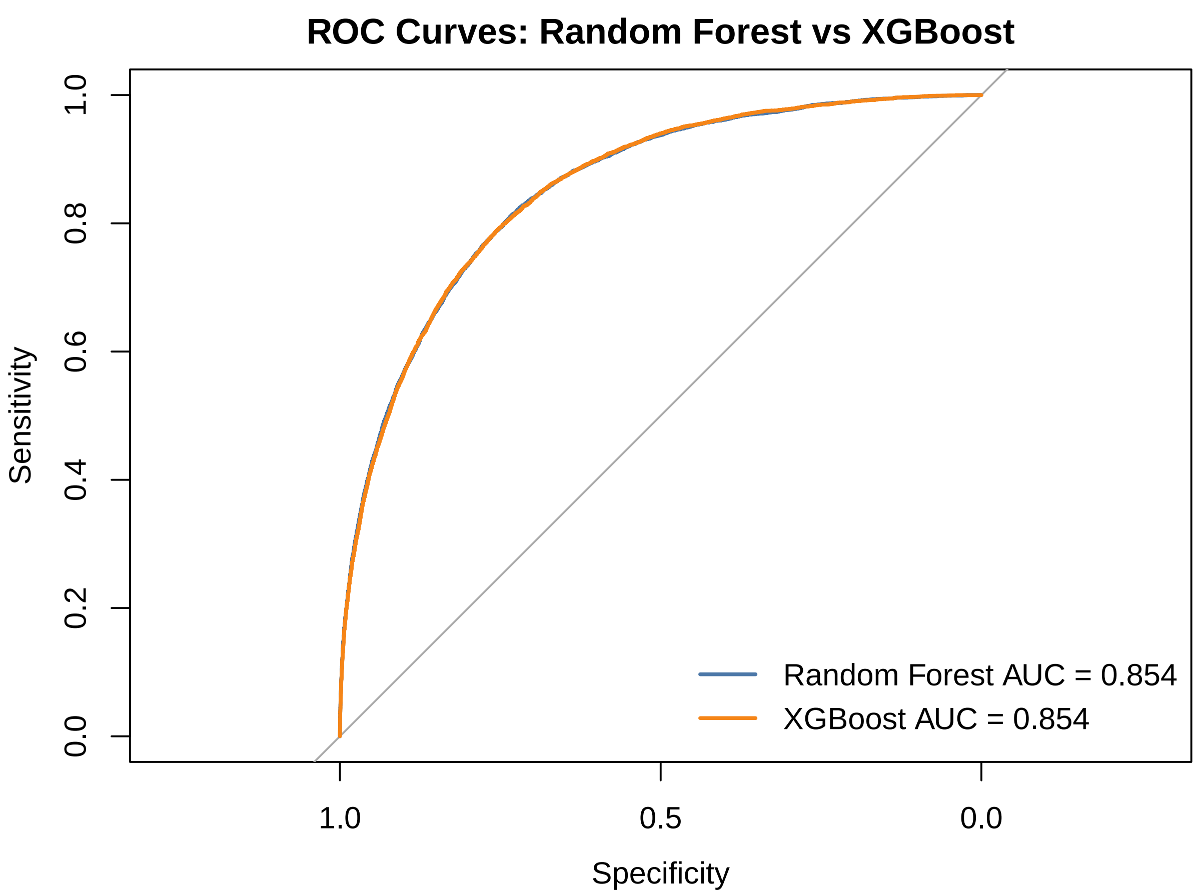
```
  xlab = "Gain",
  col = "#F58518",
  border = NA
)
```




```
par(old_par)
```

```
#png(filename = "../figures/roc_rf_xgb.png", width = 900, height = 650, res = 150)

roc_rf <- roc(y_test, pred_test_rf, quiet = TRUE)
roc_xgb <- roc(y_test, pred_test_xgb, quiet = TRUE)

plot(
  roc_rf,
  col = "#4C78A8",
  lwd = 2,
  main = "ROC Curves: Random Forest vs XGBoost"
)
plot(
  roc_xgb,
  col = "#F58518",
  lwd = 2,
```

```
  add = TRUE
)

legend(
  "bottomright",
  legend = c(
    paste0("Random Forest AUC = ", round(as.numeric(auc(roc_rf)), 3)),
    paste0("XGBoost AUC = ", round(as.numeric(auc(roc_xgb)), 3))
  ),
  col = c("#4C78A8", "#F58518"),
  lwd = 2,
  bty = "n"
)
```



Final model comparison

```r
model_comparison <- rbind(
  evaluate_model(
    "Logistic (Weighted)",
    y_train,
    pred_train_w_glm,
    y_test,
    pred_test_w_glm
  ),
  evaluate_model(
    "Logistic (Unweighted)",
    y_train,
    pred_train_u_glm,
    y_test,
    pred_test_u_glm
  ),
  evaluate_model(
    "LASSO (Weighted)",
    y_train,
    pred_train_w_lasso,
    y_test,
    pred_test_w_lasso
  ),
  evaluate_model(
    "LASSO (Unweighted)",
    y_train,
    pred_train_u_lasso,
    y_test,
    pred_test_u_lasso
  ),
  evaluate_model(
    "GAM (Weighted)",
    y_train,
    pred_train_w_gam,
    y_test,
    pred_test_w_gam
  ),
  evaluate_model(
    "GAM (Unweighted)",
    y_train,
```

```r
    pred_train_u_gam,
    y_test,
    pred_test_u_gam
  ),
  evaluate_model(
    "Random Forest",
    y_train,
    pred_train_rf,
    y_test,
    pred_test_rf
  ),
  evaluate_model(
    "XGBoost",
    y_train,
    pred_train_xgb,
    y_test,
    pred_test_xgb
  )
)

model_comparison <- model_comparison %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  arrange(desc(Test_AUC))

model_comparison
```

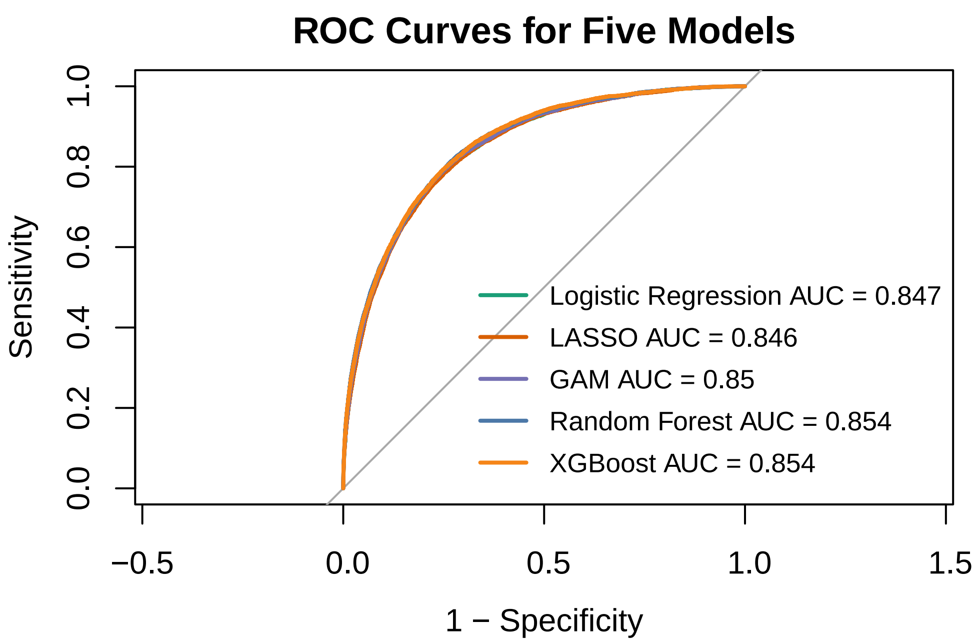
```
                   Model Test_AUC Threshold Accuracy Sensitivity Specificity
1          Random Forest    0.854     0.112    0.780       0.759       0.782
2                XGBoost    0.854     0.464    0.749       0.799       0.744
3       GAM (Unweighted)    0.850     0.078    0.746       0.799       0.740
4         GAM (Weighted)    0.849     0.087    0.758       0.785       0.756
5    Logistic (Weighted)    0.847     0.085    0.754       0.786       0.751
6  Logistic (Unweighted)    0.847     0.076    0.740       0.801       0.734
7     LASSO (Unweighted)    0.846     0.085    0.758       0.775       0.756
8       LASSO (Weighted)    0.845     0.085    0.741       0.794       0.735
```

Visuals

ROC Curves for Five Unweighted Models

```
library(pROC)

# ROC objects for five unweighted / non-survey-weighted models
roc_logistic <- roc(y_test, pred_test_u_glm, quiet = TRUE)
roc_lasso    <- roc(y_test, pred_test_u_lasso, quiet = TRUE)
roc_gam      <- roc(y_test, pred_test_u_gam, quiet = TRUE)
roc_rf       <- roc(y_test, pred_test_rf, quiet = TRUE)
roc_xgb      <- roc(y_test, pred_test_xgb, quiet = TRUE)

roc_list <- list(
  "Logistic Regression" = roc_logistic,
  "LASSO" = roc_lasso,
  "GAM" = roc_gam,
  "Random Forest" = roc_rf,
  "XGBoost" = roc_xgb
)

cols <- c("#1B9E77", "#D95F02", "#7570B3", "#4C78A8", "#F58518")

# Save ROC curve as PNG
#png("../figures/roc_curves_5_models.png", width = 900, height = 700, res = 120)

plot(
  roc_list[[1]],
  col = cols[1],
  lwd = 2,
  legacy.axes = TRUE,
  main = "ROC Curves for Five Models",
  xlab = "1 - Specificity",
  ylab = "Sensitivity"
)

for (i in 2:length(roc_list)) {
  plot(
    roc_list[[i]],
    col = cols[i],
    lwd = 2,
```

```
    add = TRUE
  )
}

legend(
  "bottomright",
  legend = paste0(
    names(roc_list),
    " AUC = ",
    round(sapply(roc_list, function(x) as.numeric(auc(x))), 3)
  ),
  col = cols,
  lwd = 2,
  cex = 0.8,
  bty = "n"
)
```




```
#dev.off()
```

Table of five unweighted models

```
library(dplyr)
library(knitr)

table_5_unweighted <- model_comparison %>%
  filter(Model %in% c(
    "Logistic (Unweighted)",
    "LASSO (Unweighted)",
    "GAM (Unweighted)",
    "Random Forest",
    "XGBoost"
  )) %>%
  arrange(desc(Test_AUC))

kable(
  table_5_unweighted,
  digits = 3,
  align = "lrrrrr"
)
```

Model	Test_AUC	Threshold	Accuracy	Sensitivity	Specificity
Random Forest	0.854	0.112	0.780	0.759	0.782
XGBoost	0.854	0.464	0.749	0.799	0.744
GAM (Unweighted)	0.850	0.078	0.746	0.799	0.740
Logistic (Unweighted)	0.847	0.076	0.740	0.801	0.734
LASSO (Unweighted)	0.846	0.085	0.758	0.775	0.756

Table of weighted versus unweighted comparison for regression-based models

```
table_weighted_vs_unweighted <- model_comparison %>%
  filter(Model %in% c(
    "Logistic (Weighted)",
    "Logistic (Unweighted)",
    "LASSO (Weighted)",
    "LASSO (Unweighted)",
```

```r
    "GAM (Weighted)",
    "GAM (Unweighted)"
  )) %>%
  mutate(
    Base_Model = case_when(
      grepl("Logistic", Model) ~ "Logistic Regression",
      grepl("LASSO", Model) ~ "LASSO",
      grepl("GAM", Model) ~ "GAM"
    ),
    Weighting = case_when(
      grepl("\\(Weighted\\)", Model) ~ "Weighted",
      grepl("\\(Unweighted\\)", Model) ~ "Unweighted"
    )
  ) %>%
  select(
    Base_Model,
    Weighting,
    Test_AUC,
    Threshold,
    Accuracy,
    Sensitivity,
    Specificity
  ) %>%
  arrange(Base_Model, Weighting)

kable(
  table_weighted_vs_unweighted,
  digits = 3,
  align = "llrrrrr"
)
```

Base_Model	Weighting	Test_AUC	Threshold	Accuracy	Sensitivity	Specificity
GAM	Unweighted	0.850	0.078	0.746	0.799	0.740
GAM	Weighted	0.849	0.087	0.758	0.785	0.756
LASSO	Unweighted	0.846	0.085	0.758	0.775	0.756
LASSO	Weighted	0.845	0.085	0.741	0.794	0.735
Logistic Regression	Unweighted	0.847	0.076	0.740	0.801	0.734
Logistic Regression	Weighted	0.847	0.085	0.754	0.786	0.751